

Bangladesh Data Analysis

Gustavo Cepparo and Milica Cudina

As before, first we import the data.

```
bangladesh=read.csv("bangladesh-data.csv", header=TRUE)
names(bangladesh)

## [1] "Arsenic" "Chlorine" "Cobalt"

#accessing single columns
#bangladesh$Arsenic
attach(bangladesh)
Arsenic

## [1] 2400 6 904 321 1280 151 141 1050 511 688 81 8 37 6 22
## [16] 43 39 92 253 200 255 1150 1180 9 107 6 149 6 46 13
## [31] 6 150 6 189 364 42 390 6 270 248 139 6 82 82 256
## [46] 165 6 180 86 6 38 262 404 8 85 98 6 22 6 6
## [61] 6 15 103 86 6 46 62 43 6 6 55 6 107 65 276
## [76] 114 6 6 6 65 142 194 6 54 702 6 986 153 84 16
## [91] 1460 306 49 36 106 6 41 84 278 41
## [ reached 'max' / getOption("max.print") -- omitted 171 entries ]

mean(Arsenic)

## [1] 125.3199

var(Arsenic)

## [1] 88789.39

#hist(Arsenic)
#what are the dimensions of `bangladesh`
dim(bangladesh)

## [1] 271 3

#see if there are missing data
bangladesh=na.omit(bangladesh)
#what are the "new" dimensions of `bangladesh`
dim(bangladesh)

## [1] 268 3

attach(bangladesh)

## The following objects are masked from bangladesh (pos = 3):
##
## Arsenic, Chlorine, Cobalt

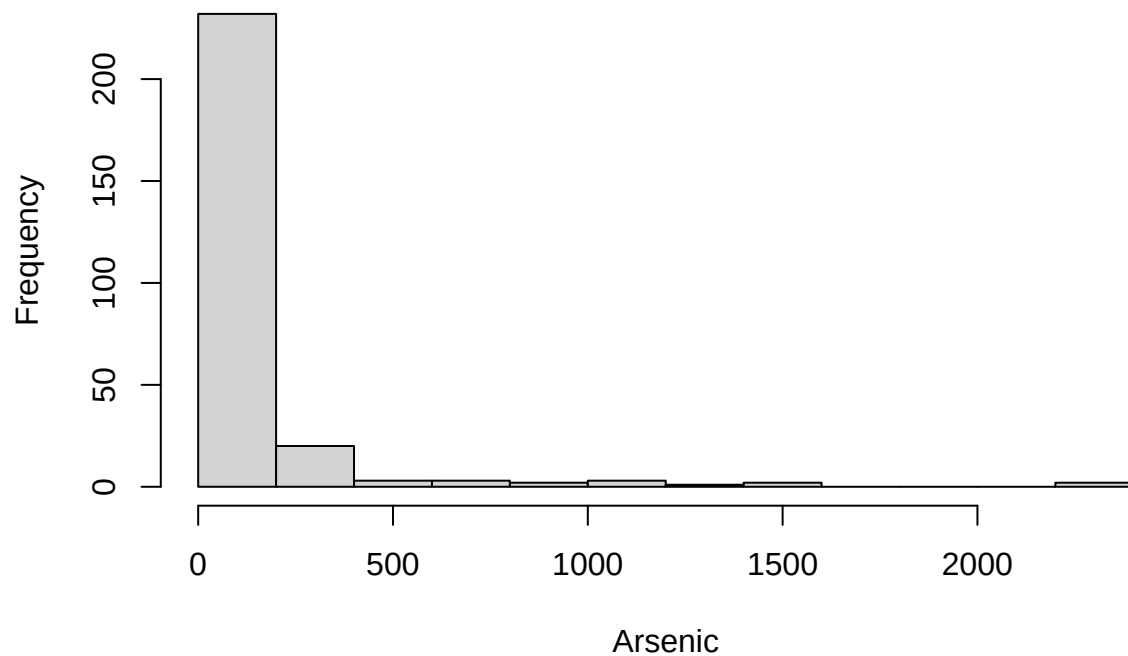
n=length(Arsenic)
n
```

```
## [1] 268
```

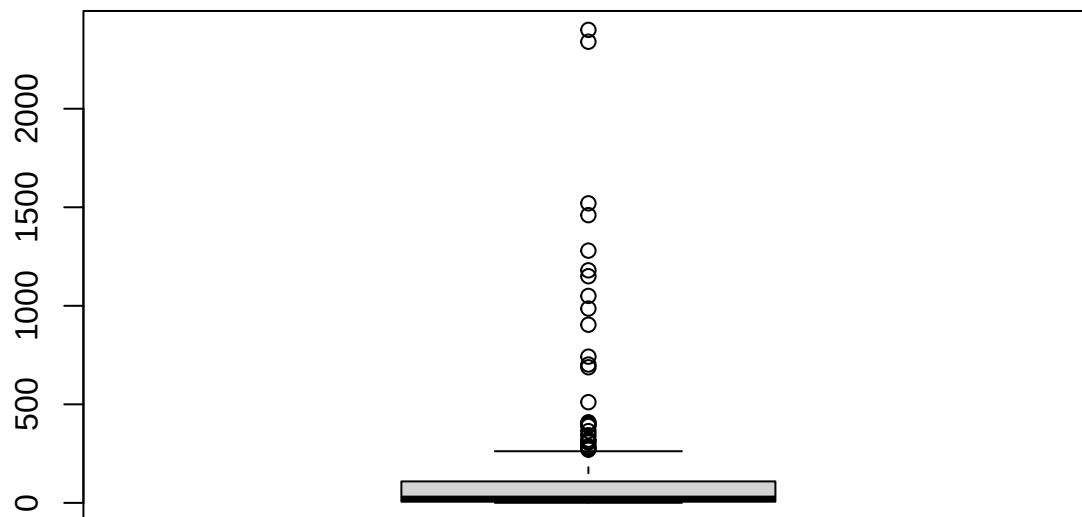
Again, we undertake a rudimentary exploratory data analysis.

```
hist(Arsenic)
```

Histogram of Arsenic



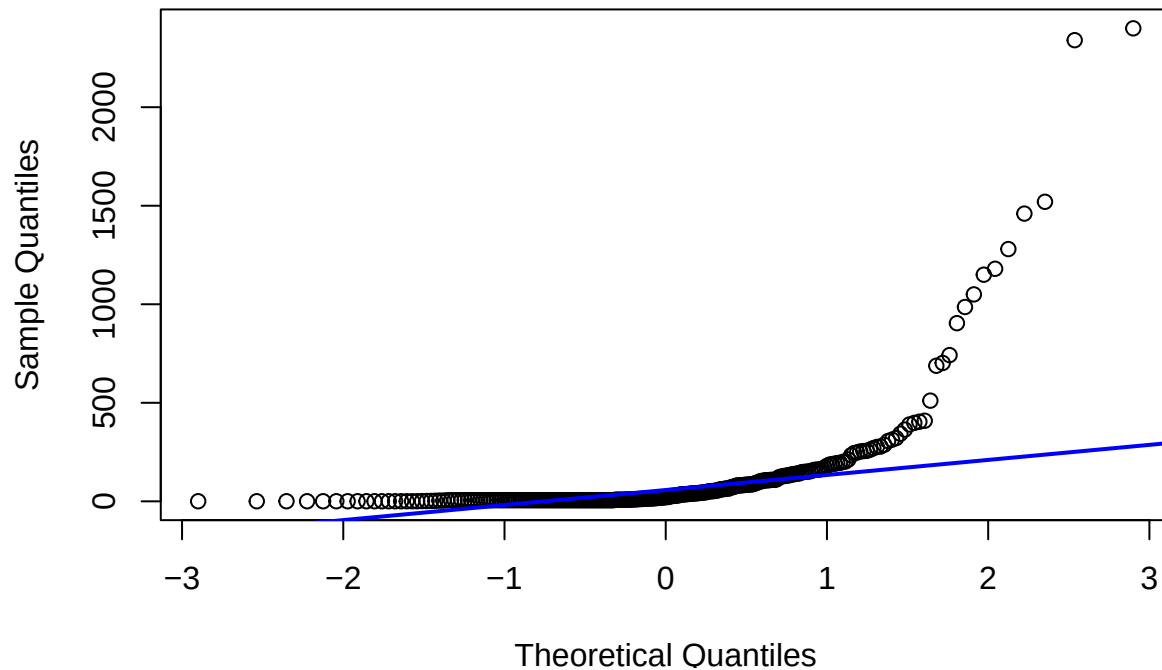
```
boxplot(Arsenic)
```



```
qqnorm(Arsenic)
```

```
qqline(Arsenic, col="blue", lwd=2)
```

Normal Q-Q Plot



What does the test of normality tell us?

```
shapiro.test(Arsenic)
```

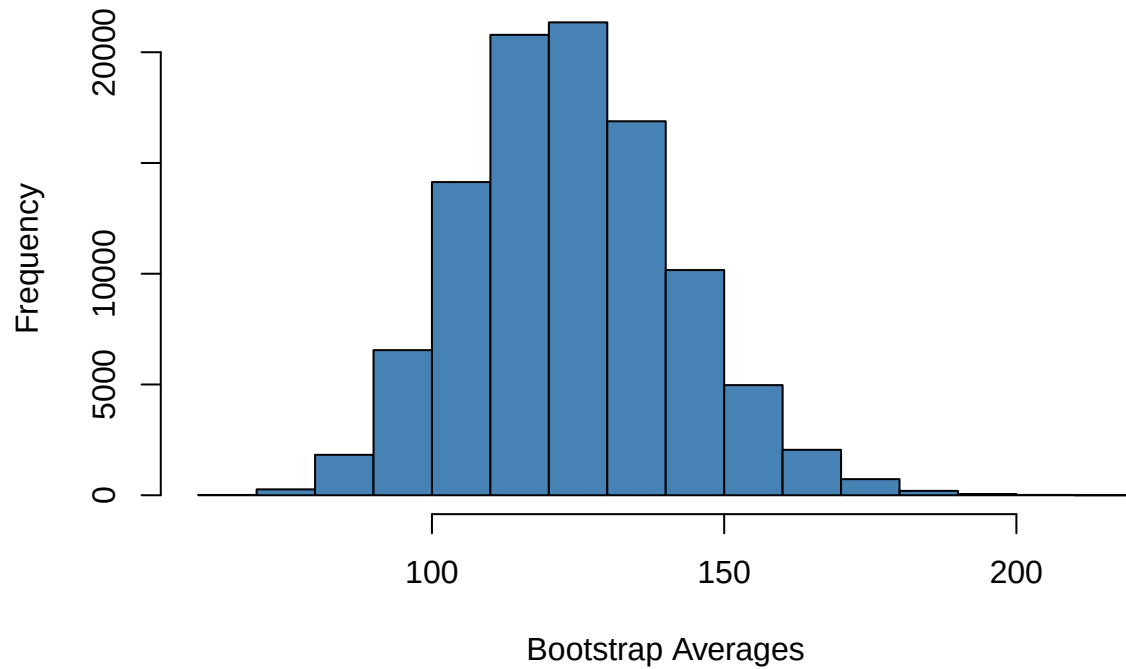
```
##  
##  Shapiro-Wilk normality test  
##  
## data:  Arsenic  
## W = 0.42284, p-value < 2.2e-16
```

Even though we have ample evidence against the normality of the data, due to the large sample size, we could use the classical approach to the confidence interval.

Now, what about bootstrap?

```
n.boot=10^5  
arsenic.mean=replicate(n.boot, mean(sample(Arsenic, n, replace=TRUE)))  
hist(arsenic.mean,  
     main="Bootstrap Distribution of Averages",  
     xlab="Bootstrap Averages",  
     col="steelblue")
```

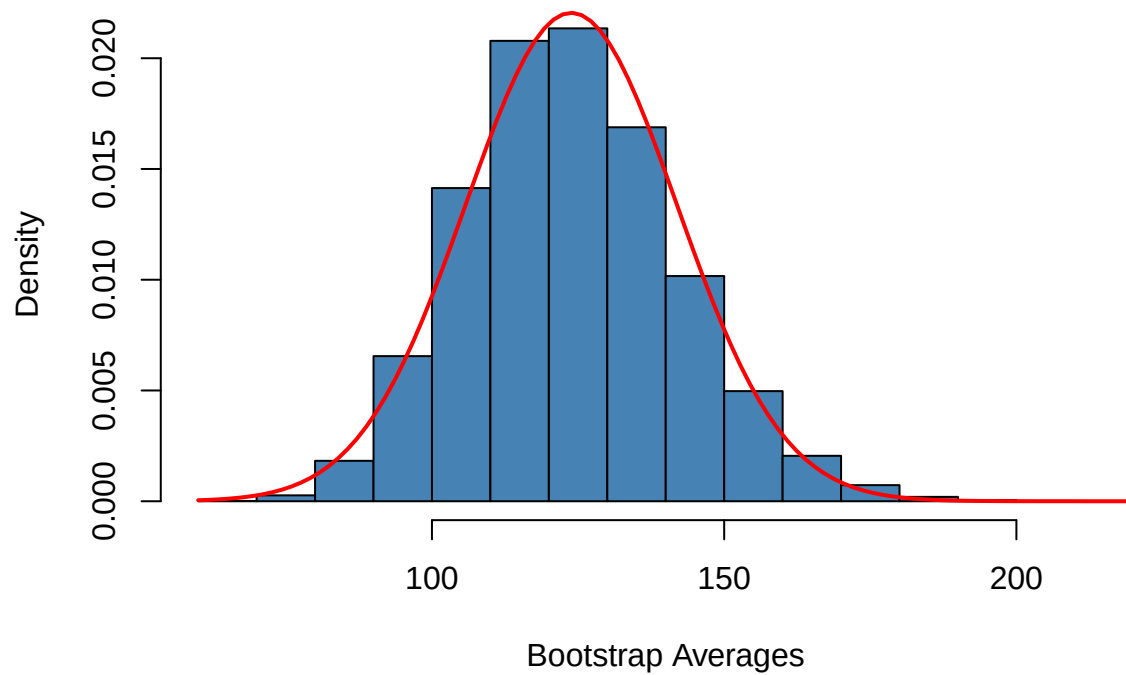
Bootstrap Distribution of Averages



Superimposing the normal bell curve.

```
hist(arsenic.mean,  
     main="Bootstrap Distribution of Averages",  
     xlab="Bootstrap Averages",  
     col="steelblue",  
     prob=TRUE)  
curve(dnorm(x, mean=mean(arsenic.mean), sd=sd(arsenic.mean)), col="red", lwd=2, add=TRUE)
```

Bootstrap Distribution of Averages



We could now construct a 2SE bootstrap confidence interval.

```
#bootstrap mean
mu.boot=mean(arsenic.mean)
mu.boot
```

```
## [1] 123.8397
```

```
#bootstrap SE
se.boot=sd(arsenic.mean)
```

```
#lower bound
l.bd=mu.boot-2*se.boot
#upper bound
u.bd=mu.boot+2*se.boot
```

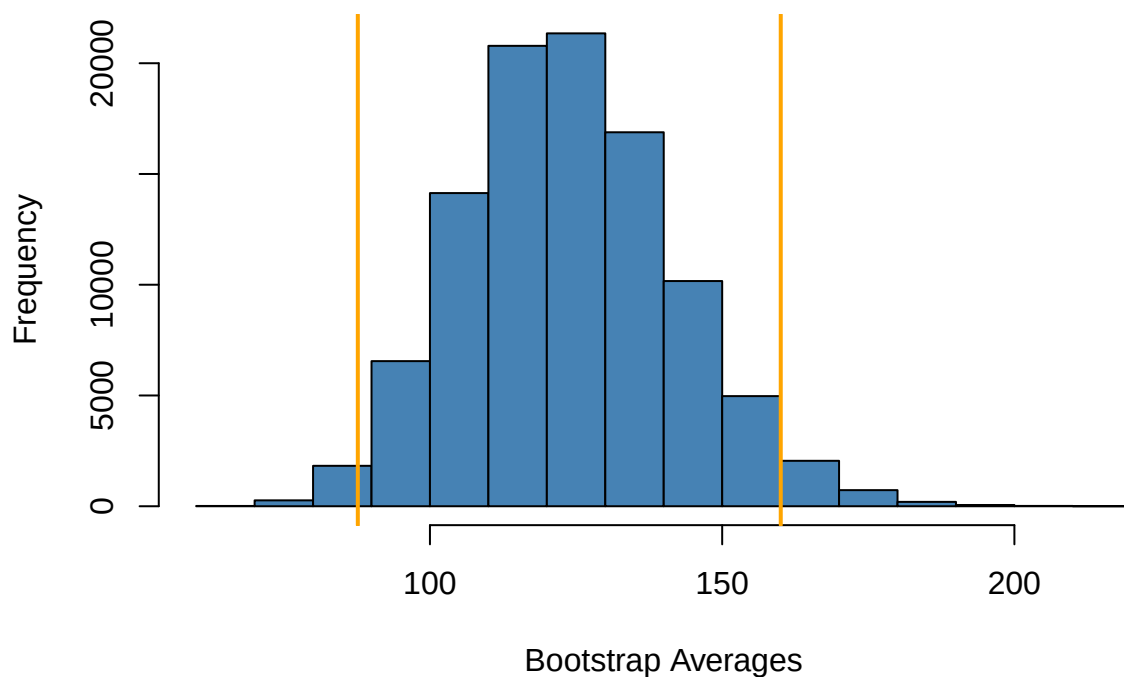
```
print(c(l.bd, u.bd))
```

```
## [1] 87.65667 160.02273
```

It might be interesting to superimpose it on the histogram.

```
hist(arsenic.mean,
     main="Bootstrap Distribution of Averages",
     xlab="Bootstrap Averages",
     col="steelblue")
abline(v=l.bd, col="orange", lwd=2)
abline(v=u.bd, col="orange", lwd=2)
```

Bootstrap Distribution of Averages



We can also construct a bootstrap 95%-percentile confidence interval.

```
median(arsenic.mean)
```

```
## [1] 122.9362
```

```
bds=quantile(arsenic.mean, c(0.025, 0.975))
```

```
bds
```

```
##      2.5%      97.5%
```

```
## 91.01116 161.96908
```

An analogous plot.

```
hist(arsenic.mean,  
     main="Bootstrap Distribution of Averages",  
     xlab="Bootstrap Averages",  
     col="steelblue")  
abline(v=bds, col="orange", lwd=2)
```

Bootstrap Distribution of Averages

